

OOPs, DSA & System Design Blueprint

About ME

Hey, I'm Shubham, 7+ years in tech, 100+ interviews on both sides of the table.

Career path: Infosys → Pubmatic → Amazon → Coursera (currently WFH)

I've cracked multiple high-paying offers across India and internationally using the exact playbook below.

Connect here - [Linkedin](#) , [Instagram](#), [Whatsapp](#)

DSA Your Interview Superpower

Real talk: Miss this, miss everything. 90% of tech rounds = DSA rounds.

Master these:

- Fundamentals: Arrays, Strings, LinkedList, Stack, Queue
- Advanced: Trees, Graphs, DP, Recursion, Backtracking
- Must-know: Sliding Window, Two Pointers, Binary Search patterns

What

Where

Learn [Striver A2Z Sheet](#)

Grind **NeetCode 150** → LeetCode → Codeforces

Pro move: This sheet is more than enough, Don't memorize solutions. Learn the pattern once, apply everywhere.

Interview ammo: [Leetcode 150Questions](#) + [Blind 75](#)

2. OOP — The "Can You Code?" Test

Why: Every single language uses this. Skip it = instant red flag.

Core concepts:

- The Big 4: Encapsulation, Inheritance, Polymorphism, Abstraction
- Bonus: SOLID principles (makes you look senior)

What	Where
Theory	Apna College / Kunal Kushwaha Java
Practice	Code real examples in your preferred language
Javascript	Namaste Javascript

3. Top 80 LeetCode DSA Questions

Arrays & Hashing

#	Problem	Difficulty	Link
1	Two Sum	Easy	https://leetcode.com/problems/two-sum/
2	Contains Duplicate	Easy	https://leetcode.com/problems/contains-duplicate/
3	Valid Anagram	Easy	https://leetcode.com/problems/valid-anagram/
4	Group Anagrams	Medium	https://leetcode.com/problems/group-anagrams/

5	Top K Frequent Elements	Medium	https://leetcode.com/problems/top-k-frequent-elements/
6	Product of Array Except Self	Medium	https://leetcode.com/problems/product-of-array-except-self/
7	Encode and Decode Strings	Medium	https://leetcode.com/problems/encode-and-decode-strings/
8	Longest Consecutive Sequence	Medium	https://leetcode.com/problems/longest-consecutive-sequence/

Two Pointers

#	Problem	Difficulty	Link
9	Valid Palindrome	Easy	https://leetcode.com/problems/valid-palindrome/
10	3Sum	Medium	https://leetcode.com/problems/3sum/
11	Container With Most Water	Medium	https://leetcode.com/problems/container-with-most-water/

Sliding Window

#	Problem	Difficulty	Link
12	Best Time to Buy and Sell Stock	Easy	https://leetcode.com/problems/best-time-to-buy-and-sell-stock/
13	Longest Substring Without Repeating Characters	Medium	https://leetcode.com/problems/longest-substring-without-repeating-characters/
14	Longest Repeating Character Replacement	Medium	https://leetcode.com/problems/longest-repeating-character-replacement/
15	Minimum Window Substring	Hard	https://leetcode.com/problems/minimum-window-substring/

Stack

#	Problem	Difficulty	Link
16	Valid Parentheses	Easy	https://leetcode.com/problems/valid-parentheses/

Binary Search

#	Problem	Difficulty	Link
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17	Search in Rotated Sorted Array	Medium	https://leetcode.com/problems/search-in-rotated-sorted-array/
18	Find Minimum in Rotated Sorted Array	Medium	https://leetcode.com/problems/find-minimum-in-rotated-sorted-array/

Linked List

#	Problem	Difficulty	Link
19	Reverse Linked List	Easy	https://leetcode.com/problems/reverse-linked-list/
20	Linked List Cycle	Easy	https://leetcode.com/problems/linked-list-cycle/
21	Merge Two Sorted Lists	Easy	https://leetcode.com/problems/merge-two-sorted-lists/
22	Remove Nth Node From End of List	Medium	https://leetcode.com/problems/remove-nth-node-from-end-of-list/
23	Reorder List	Medium	https://leetcode.com/problems/reorder-list/
24	Merge K Sorted Lists	Hard	https://leetcode.com/problems/merge-k-sorted-lists/

Trees

#	Problem	Difficulty	Link
25	Invert Binary Tree	Easy	https://leetcode.com/problems/invert-binary-tree/
26	Maximum Depth of Binary Tree	Easy	https://leetcode.com/problems/maximum-depth-of-binary-tree/
27	Same Tree	Easy	https://leetcode.com/problems/same-tree/
28	Subtree of Another Tree	Easy	https://leetcode.com/problems/subtree-of-another-tree/
29	Lowest Common Ancestor of a BST	Medium	https://leetcode.com/problems/lowest-common-ancestor-of-a-binary-search-tree/
30	Binary Tree Level Order Traversal	Medium	https://leetcode.com/problems/binary-tree-level-order-traversal/
31	Validate Binary Search Tree	Medium	https://leetcode.com/problems/validate-binary-search-tree/
32	Kth Smallest Element in a BST	Medium	https://leetcode.com/problems/kth-smallest-element-in-a-bst/
33	Construct Binary Tree from Preorder and Inorder Traversal	Medium	https://leetcode.com/problems/construct-binary-tree-from-preorder-and-inorder-traversal/

34	Binary Tree Maximum Path Sum	Hard	https://leetcode.com/problems/binary-tree-maximum-path-sum/
35	Serialize and Deserialize Binary Tree	Hard	https://leetcode.com/problems/serialize-and-deserialize-binary-tree/

Tries

#	Problem	Difficulty	Link
36	Implement Trie (Prefix Tree)	Medium	https://leetcode.com/problems/implement-trie-prefix-tree/
37	Design Add and Search Words Data Structure	Medium	https://leetcode.com/problems/design-add-and-search-words-data-structure/
38	Word Search II	Hard	https://leetcode.com/problems/word-search-ii/

Heap / Priority Queue

#	Problem	Difficulty	Link
39	Find Median from Data Stream	Hard	https://leetcode.com/problems/find-median-from-data-stream/

Backtracking

#	Problem	Difficulty	Link
40	Combination Sum	Medium	https://leetcode.com/problems/combination-sum/
41	Word Search	Medium	https://leetcode.com/problems/word-search/

Graphs

#	Problem	Difficulty	Link
42	Number of Islands	Medium	https://leetcode.com/problems/number-of-islands/
43	Clone Graph	Medium	https://leetcode.com/problems/clone-graph/
44	Pacific Atlantic Water Flow	Medium	https://leetcode.com/problems/pacific-atlantic-water-flow/
45	Course Schedule	Medium	https://leetcode.com/problems/course-schedule/
46	Graph Valid Tree	Medium	https://leetcode.com/problems/graph-valid-tree/

47	Number of Connected Components in an Undirected Graph	Medium	https://leetcode.com/problems/number-of-connected-components-in-an-undirected-graph/
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48	Alien Dictionary	Hard	https://leetcode.com/problems/alien-dictionary/
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Pro tip: Graph problems boil down to: BFS, DFS, Topological Sort, or Union-Find. Learn to identify which one applies.

Dynamic Programming

#	Problem	Difficulty	Link
49	Climbing Stairs	Easy	https://leetcode.com/problems/climbing-stairs/
50	House Robber	Medium	https://leetcode.com/problems/house-robber/
51	House Robber II	Medium	https://leetcode.com/problems/house-robber-ii/
52	Longest Increasing Subsequence	Medium	https://leetcode.com/problems/longest-increasing-subsequence/
53	Coin Change	Medium	https://leetcode.com/problems/coin-change/
54	Word Break	Medium	https://leetcode.com/problems/word-break/
55	Decode Ways	Medium	https://leetcode.com/problems/decode-ways/

56	Unique Paths	Medium	https://leetcode.com/problems/unique-paths/
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57	Jump Game	Medium	https://leetcode.com/problems/jump-game/
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Secret weapon: DP is the #1 topic people fail on. Start with Climbing Stairs, understand memoization vs tabulation, then work your way up. Don't memorize — understand the state transition.

Greedy

#	Problem	Difficulty	Link
58	Maximum Subarray	Medium	https://leetcode.com/problems/maximum-subarray/
59	Jump Game	Medium	https://leetcode.com/problems/jump-game/

Intervals

#	Problem	Difficulty	Link
60	Insert Interval	Medium	https://leetcode.com/problems/insert-interval/
61	Merge Intervals	Medium	https://leetcode.com/problems/merge-intervals/

62	Non-overlapping Intervals	Medium	https://leetcode.com/problems/non-overlapping-intervals/
63	Meeting Rooms	Easy	https://leetcode.com/problems/meeting-rooms/
64	Meeting Rooms II	Medium	https://leetcode.com/problems/meeting-rooms-ii/

Math & Bit Manipulation

#	Problem	Difficulty	Link
65	Number of 1 Bits	Easy	https://leetcode.com/problems/number-of-1-bits/
66	Counting Bits	Easy	https://leetcode.com/problems/counting-bits/
67	Reverse Bits	Easy	https://leetcode.com/problems/reverse-bits/
68	Missing Number	Easy	https://leetcode.com/problems/missing-number/
69	Sum of Two Integers	Medium	https://leetcode.com/problems/sum-of-two-integers/

Strings (Additional)

#	Problem	Difficulty	Link
70	Longest Palindromic Substring	Medium	https://leetcode.com/problems/longest-palindromic-substring/
71	Palindromic Substrings	Medium	https://leetcode.com/problems/palindromic-substrings/
72	Maximum Product Subarray	Medium	https://leetcode.com/problems/maximum-product-subarray/

Matrix

#	Problem	Difficulty	Link
73	Set Matrix Zeroes	Medium	https://leetcode.com/problems/set-matrix-zeroes/
74	Spiral Matrix	Medium	https://leetcode.com/problems/spiral-matrix/
75	Rotate Image	Medium	https://leetcode.com/problems/rotate-image/

Top 30 System Design Questions

System Design is what separates SDE-1 from SDE-2/3. If DSA gets you the interview, System Design gets you the offer (and the level). No shortcuts here — you need to understand trade-offs.

Resources: Grokking the System Design Interview (Educative), System Design Primer (GitHub), Alex Xu's System Design Interview books.

Start every answer with:

- (1) Clarify requirements
- (2) Estimate scale
- (3) High-level design
- (4) Deep dive into components
- (5) Discuss trade-offs.

Easy (Start Here)

#	Problem	Key Concepts	Resource
1	Design a URL Shortener (TinyURL)	Hashing, Base62 encoding, NoSQL	Educative / Alex Xu Ch. 8
2	Design a Rate Limiter	Token bucket, Sliding window, Redis	Educative / Alex Xu Ch. 4
3	Design a Key-Value Store	Consistent hashing, Replication, CAP theorem	Alex Xu Ch. 6
4	Design Pastebin	Object storage, CDN, Expiry	Educative

5	Design a Distributed Cache	LRU eviction, Cache invalidation, Redis/Memcached	System Design Primer
6	Design a Content Delivery Network (CDN)	Edge servers, Cache policies, DNS routing	Alex Xu Ch. 10
7	Design a Web Crawler	BFS crawling, URL frontier, Deduplication	Alex Xu Ch. 9
8	Design an Authentication System (SSO)	OAuth 2.0, JWT, SAML, Session management	Educative
9	Design a Pub/Sub System (like Kafka)	Message queues, Topics, Consumer groups	Educative
10	Design a Notification System	Push/Pull, Fan-out, Priority queues	Alex Xu Ch. 10

Medium (Core Interview Questions)

#	Problem	Key Concepts	Resource
11	Design Instagram	News feed, Image storage, CDN, Sharding	Educative / Alex Xu
12	Design Twitter / X	Fan-out on write vs read, Timeline, Celebrity problem	Grokking SDI

13	Design WhatsApp / Chat System	WebSockets, Message queues, Presence, E2E encryption	Alex Xu Ch. 12
14	Design YouTube / Netflix	Video transcoding, Adaptive streaming, CDN	Alex Xu Ch. 14
15	Design Uber / Lyft	Geospatial indexing, Matching, ETA, Real-time tracking	Educative
16	Design Dropbox / Google Drive	File chunking, Sync, Deduplication, Conflict resolution	Alex Xu Ch. 15
17	Design a Search Autocomplete / Typeahead	Trie, Prefix matching, Caching, Ranking	Alex Xu Ch. 13
18	Design a News Feed System	Fan-out, Ranking algorithm, Caching, Pagination	Alex Xu Ch. 11
19	Design an E-Commerce Platform (Amazon)	Microservices, Inventory, Payment, Search	Educative
20	Design Yelp / Nearby Search	Geohashing, QuadTree, Proximity search	Alex Xu Vol. 2
21	Design a Ticket Booking System (BookMyShow)	Seat locking, Distributed transactions, Concurrency	Educative
22	Design a Task Scheduler	Priority queues, Cron, Distributed locking	System Design Primer

Secret weapon: Interviewers care more about trade-offs than the "right" answer. Always say:
"We could do X which gives us Y, but the trade-off is Z."

Hard (Staff / Senior Level)

#	Problem	Key Concepts	Resource
23	Design Google Maps	Graph algorithms, Tile rendering, Routing, Caching	Alex Xu Vol. 2
24	Design a Distributed Message Queue (Kafka)	Partitioning, Replication, Consumer groups, Ordering	Educative
25	Design Google Docs (Real-time Collaboration)	CRDT / OT, WebSockets, Conflict resolution	Educative
26	Design a Search Engine (Google Search)	Inverted index, PageRank, Crawling, Sharding	System Design Primer
27	Design a Payment System (Stripe)	Idempotency, Double-entry ledger, Reconciliation	Alex Xu Vol. 2
28	Design a Stock Exchange	Order matching engine, Low latency, FIFO queues	Educative
29	Design a Distributed File System (HDFS / GFS)	Chunk servers, Master node, Replication, Fault tolerance	Research papers

Must-Know System Design Concepts

Before tackling the 30 questions above, make sure you're solid on these fundamentals:

Scaling: Horizontal vs Vertical scaling, Load Balancing (Round Robin, Least Connections, Consistent Hashing)

Databases: SQL vs NoSQL, Sharding, Replication (Master-Slave, Master-Master), ACID vs BASE, Indexing

Caching: Cache-aside, Write-through, Write-back, CDN caching, Redis/Memcached

Networking: REST vs gRPC, WebSockets, Long Polling, DNS, API Gateway

Reliability: CAP Theorem, Eventual Consistency, Circuit Breakers, Retry with Backoff, Idempotency

Storage: Blob storage, Object storage, Block storage, Data lakes

The 8-Week Execution Plan

Weeks 1-2: Arrays, Hashing, Two Pointers, Sliding Window (20 problems, 2 hrs/day)

Weeks 3-4: Trees, Graphs, Linked Lists (20 problems, 2 hrs/day)

Weeks 5-6: DP, Backtracking, Greedy, Intervals (20 problems, 2 hrs/day)

Weeks 7-8: Remaining 15 problems + System Design (2 SD questions/week, 3 hrs/day)

Throughout: Revise solved problems every weekend. If you can't re-solve in 15 min, you didn't learn it.

Final Words

DSA isn't about memorizing 500 problems. It's about recognizing 15 patterns and applying them fast.

The brutal truth: Solving without understanding = wasted time. Understanding without practice = no speed. Speed without mock interviews = choking on the day.

Your action plan: Pick problem #1 from each category. Start TODAY. Track your progress. Do at least 2 problems/day. Mock interview weekly.

The market rewards prepared engineers. Most people star the Blind 75 list and never open it again. Don't be most people.

Your Interview Journey Starts Now.

Happy Learning :)